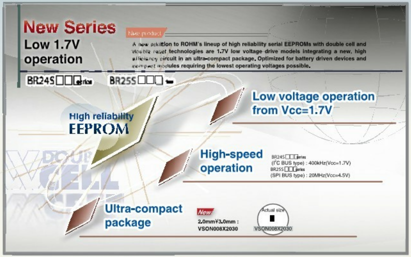
ROHM's High Reliability Series Serial EEPROMs Double Cell • Double Reset Method

Lineup

High reliability series

Serial EEPROMs

ROHM's Serial EEPROMs represent the utmost in reliability.
The lineup is available in the three standard interfaces for compatibility with virtually any application.



New Series Low 1.7V operation

New product

A new addition to ROHM's lineup of high reliability serial EEPROMs with double cell and double reset technologies are 1.7V low voltage drive models integrating a new, high efficiency circuit in an ultra-compact package. Optimized for battery driven devices and compact modules requiring the lowest operating voltages possible.

BR24S□□□series BR25S□□□series

High reliability EEPROM

Low voltage operation from Vcc=1.7V

High-speed operation

BR24S□□□series
(I²C BUS type) : 400kHz(Vcc=1.7V)
BR25S□□□series
(SPI BUS type) : 20MHz(Vcc=4.5V)

Ultra-compact package

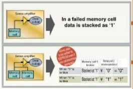
New
2.0mm¥3.0mm :
VSON008X2030

Actual size
VSON008X2030

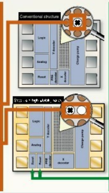
The reasons behind ROHM's unparalleled reliability.

Double cell

Data is written to the EEPROM by passing electrons through a tunnel-oxide film. This process places a significant amount of stress on the film, possibly leading to erosion and, subsequently memory bit failures that can prevent data from being written to the cells. To avoid this, ROHM utilizes a unique double cell structure that allocates two cells for each bit of data. Therefore, if one cell malfunctions the other will continue to work independently. 3



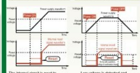
In a failed memory cell data is stacked as '1'



Double reset

ICs are generally robust against power interruptions (i.e. ON/OFF) and will normally recover. EEPROMs, on the other hand, are more susceptible, and may malfunction due to unstable power conditions. In order to combat this ROHM integrates two protection circuits, a Power ON Reset and a Vcc Lockout circuit, that detect power interruptions and low voltage conditions during power ON/OFF and resets the internal circuitry, preventing writing errors.

Power ON Reset (POR) Function　Vcc-Lock Out (LVCC) Function



The internal circuit is reset to prevent unstable situations that can cause EEPROM malfunctions during power ON.

Low voltage is detected and the internal circuitry reset to prevent malfunction.

I²C BUS

BR24□□□□family
- 2-wire communication
- Simple system
- 8bit operation

Supply voltage 1.8V to 5.5V, Operating temperature −40°C to +85°C
BR24L□□series — DOUBLE CELL · DOUBLE RESET · A/Au · 1.8V~5.5V · -40°C +85°C · WRITE 5ms · PAGE WRITE

New
Supply voltage 1.7V to 5.5V, Operating temperature −40°C to +85°C
BR24S□□□series — DOUBLE CELL · DOUBLE RESET · A/Au · 1.7V~5.5V · -40°C +85°C · WRITE 5ms · PAGE WRITE · 400kHz · 4kV

Supply voltage 2.5V to 5.5V, Operating temperature −40°C to +125°C
☆BR24H□□series — DOUBLE CELL · DOUBLE RESET · A/Au · 2.5V~5.5V · -40°C +125°C · WRITE 5ms · PAGE WRITE · 6kV

☆: Under development

SPI BUS

BR25□□□□family
- High-speed communication 20MHz
- Data-protection function (software / hardware area designation)
- 8bit operation

Supply voltage 1.8V to 5.5V, Operating temperature −40°C to +85°C
BR25L□□0series — DOUBLE CELL · DOUBLE RESET · A/Au · 1.8V~5.5V · -40°C +85°C · WRITE 5ms · PAGE WRITE · STATUS REGISTER

New
Supply voltage 1.7V to 5.5V, Operating temperature −40°C to +85°C
BR25S□□□series — DOUBLE CELL · DOUBLE RESET · A/Au · 1.7V~5.5V · -40°C +85°C · WRITE 5ms · PAGE WRITE · STATUS REGISTER · 20MHz · 4kV

Supply voltage 2.5V to 5.5V, Operating temperature −40°C to +125°C
BR25H□□0series — DOUBLE CELL · DOUBLE RESET · A/Au · 2.5V~5.5V · -40°C +125°C · WRITE 5ms · PAGE WRITE · STATUS REGISTER · 6kV

Microwire BUS

BR93□□□family
- High speed communication 2MHz
- Origin serial EEPROMs
- 16bit operation

Supply voltage 1.8V to 5.5V, Operating temperature −40°C to +85°C
BR93L□□series — DOUBLE CELL · DOUBLE RESET · A/Au · 1.8V~5.5V · -40°C +85°C · WRITE 5ms · WRITE ALL · READY BUSY

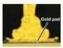
Supply voltage 2.5V to 5.5V, Operating temperature −40°C to +105°C
BR93A□□series — DOUBLE CELL · DOUBLE RESET · A/Au · 2.5V~5.5V · -40°C +105°C · WRITE 5ms · WRITE ALL · READY BUSY

Supply voltage 2.7V to 5.5V, Operating temperature −40°C to +125°C
BR93H□□series — DOUBLE CELL · DOUBLE RESET · A/Au · 2.7V~5.5V · -40°C +125°C · WRITE 10ms · WRITE ALL · READY BUSY · 8kV

Gold to Gold connection

ROHM uses gold (Au) pads instead of aluminum and gold wires for same-metal connection, improving reliability significantly over conventional products.

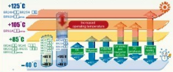
Gold Pad



Gold Pad Junction Cross Section

+125°C operation

Sophisticated circuit design and process technology are utilized to widen the operating temperature range, from −40°C to +125°C. This makes them suitable for a wide range of applications, particularly those requiring a greater degree of reliability.



Explanation of Icons	
-40°C +85°C · -40°C +105°C · -40°C +125°C	Operating temperature range
DOUBLE CELL	Double cell
DOUBLE RESET	Double reset
A/Au	Gold pad/Gold wire connections used for greater reliability
1.7V~5.5V · 1.8V~5.5V · 2.5V~5.5V · 2.7V~5.5V	Supply voltage range
-40°C +85°C · -40°C +105°C · -40°C +125°C	Operating temperature range
WRITE 5ms · WRITE 10ms	Write cycle time
WRITE ALL	Equipped with Write All function
READY BUSY	Ready/Busy status display function included
PAGE WRITE	Equipped with Page Write function
STATUS REGISTER	Built-in status register
400kHz · 20MHz	Maximum operating frequency
4kV · 6kV · 8kV	4/6/8kV ESD Resistance

General Purpose ICs
Video and Imaging ICs
Audio ICs
IT Equipment ICs
Consumer Product ICs
Automotive ICs
Discrete Semiconductors
Passive Components
Optoelectronics
Modules

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